# Crop Maturity Detection using Deep Learning and IoT

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***Abstract*—Crop maturity detection plays a crucial role in increasing agricultural productivity and product value. However, current methods of manual inspection suffer from inaccuracies due to variations in light intensity and inspectors' fatigue. Therefore, this paper adopts deep learning to automatically classify crop maturity in tomatoes and dragon fruits using image processing. To achieve optimal performance, four deep learning architectures were tested using transfer learning techniques: EfficientNetV2-B0, MobileNetV2, ConvNeXt-Tiny, and MicroViT-S1. The ConvNeXt-Tiny architecture achieved the highest performance for both types of crops, providing 100% accuracy on all evaluation metrics. For application purposes, tomato-based algorithms were fine-tuned and deployed to the Scientech 6205AI edge platform. It was found that the MicroViT-S1 algorithm gives the lowest latency rate among other models with 61.73 FPS and a mean latency of 16.14 ms. Based on this analysis, ConvNeXt-Tiny delivers the best accuracy rates, whereas MicroViT-S1 is more efficient and practical for real-life scenarios.**

***Index Terms*—Crop maturity detection, tomato, dragon fruit, deep learning, edge device, real-time inference, image classification.**

## I. Introduction

Maturity Detection is an important part of Agriculture as it affects product quality, market value, and overall yield. Incorrect timing can lead to reduced profit and wastage. Therefore, harvesting the crop at the proper time is crucial.

We can also manually identify maturity, but manual detection depends on what your eyes see and how much you know. Also, it becomes tricky when some fruits look almost the same when nearly ripe. That's why manually checking is not always trustworthy, especially when things come to consistent and fast maturity detection. Additionally, it takes lots of manpower for large agricultural fields.

To overcome this problem, computer vision provides a solution. Computer vision is a fast, safe and reliable way to check the quality of crops without damaging them. It helps in automatic inspection and sorting of crops, reducing human errors, and saving time. In crop maturity detection, a computer vision-based method is an affordable and effective replacement for manual checking.

Deep Learning combined with computer vision has greatly enhanced the classification process using images. One example is CNN, which has the capability of automatically detecting important patterns and features from images and hence resulting in better performance compared to other traditional learning algorithms. The downside is that deep learning models take much effort to train since much data and computing power are needed. This challenge can be addressed using Transfer Learning in which a model is first trained using ImageNet before it can be modified for specific uses like crop maturity classification.

There have been several pretrained models used in different studies for the purpose of classifying maturity of tomatoes. For instance, Begum et al. [2] explored the use of transfer learning techniques using VGG, Inception, and ResNet to classify images of tomatoes into immature, partly mature, and fully mature, with the VGG 19 model giving the best classification result of 97.37%. Appe et al. [3] developed a deep model consisting of VGG16 and Multi-layer Perceptron (MLP), which was able to classify the ripeness of the tomato based on 700 ripe and 700 unripe images, resulting in precision of 0.96 and recall of 0.97.

Also, Mu et al. [4] have developed tomato detection model based on the RCNN with architectural types like ResNet 50, ResNet 101, and Inception-Resnet-v2, whereby R-CNN+ResNet 101 achieved the highest average precision of 87.83. Furthermore, Mputu et al. [6] have developed a hybrid model that involves use of CNN models pretrained and include MobileNetV2, Inceptionv3, AlexNet, and Resnet50, thus achieving accuracy of 97.50% in binary classes and 96.67% in multiclass.

In one of their studies, Hoang-Tu et al. [7] applied the YOLOv9 algorithm for ripeness level classification and tomato counting, categorizing the tomato into three categories: Fully-ripe, Semi-ripe, and Unripe. In this project, 2610 images were used. Similarly, Khan et al. [9] introduced a modular convolutional transformer-based model that was assessed against Seg.

Nikam et al. [12] presented a proposed system based on deep CNN feature fusion and ensemble learning method for the purpose of predicting maturity levels and assessing the quality grades of dragon fruit. This system consists of combining the features extracted from several CNNs and passing through several classifiers, which helps to improve prediction efficiency, robustness, and generalization capability.

In another research, Aishwarya et al. [13] presented a framework based on pre-trained deep learning models such as VGG16, MobileNetV3, and EfficientNetB0 for the automatic detection of maturity levels and quality grade classification of dragon fruits. Several activation functions, ELU, ReLU, and SELU, were utilized to capture feature information, which was subsequently classified using SVM, Decision Tree, Random Forest, AdaBoost, and CatBoost classifiers. The highest accuracy was attained using EfficientNetB0-SVM and MobileNetV3-SVM models, with 85% and 99% accuracy, respectively.

On the other hand, Khatun et al. [14] created a real-world dataset for identifying ripeness level and assessing the quality grades of dragon fruits using image acquisition in four months at three locations in Bangladesh. The images were taken with the assistance of domain experts. Immature, Mature, Fresh, and Defective fruit classes were involved in this study, captured under diverse lighting conditions, backgrounds, and fruit states.

In this work, four deep learning models; EfficientNetV2B0 [12], MobileNetV2 [13], ConvNeXt-Tiny [14], and MicroViT-S1 [15], classify tomatoes and dragon fruits into various maturity classes. These models have different advantages concerning the accuracy, efficiency of processing, and the complexity of the models. This paper seeks to identify the most efficient model for classifying crops maturity using image data and transfer learning.

This research paper is structured into five chapters, namely: Chapter II Implementation details, Chapter III Results, Chapter IV Conclusion and future works, and Chapter V Acknowledgment.

## II. Implementaion Details

### A. Dataset and Preprocessing

*1) Tomato Dataset:* Tomato, a widely cultivated fruit plant, plays an essential role in both nutrition and economy. In this study, a custom-collected dataset was acquired from a local vegetable market in Baramati, Maharashtra, during 2026. The data collection focused on capturing the visual progression of tomatoes across five distinct maturity stages.

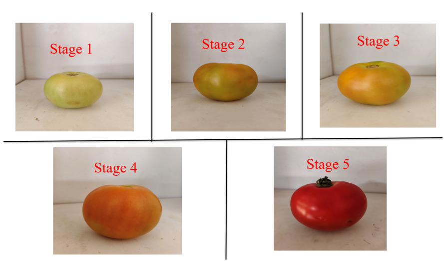


Fig. 1: Sample images from the tomato dataset.

Twenty high-quality images per each stage were obtained, thus forming the initial set of 100 images. It should be noted that all images were taken with a Vivo V40 mobile phone with a 48-megapixel camera to ensure maximum sharpness and quality. The images are saved in JPG format, having the size of $800 \times 800$ pixels.

To enrich the dataset and avoid overfitting, various image augmentation operations were applied, namely, rotations, CLAHE, hue-saturation changes, flipping horizontally and vertically, and enhancing brightness contrast. As a result, the augmented dataset became 800 images large.

Finally, the dataset was split into the training, validation, and test sets containing 560, 120, and 120 images correspondingly.

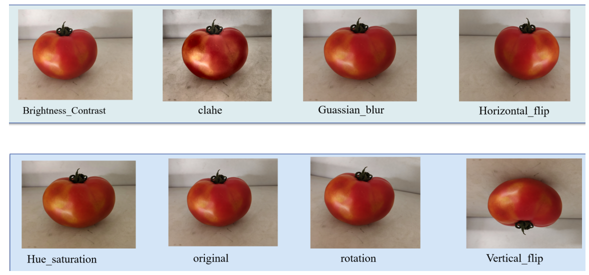


Fig. 2: Augumentation of images.

TABLE I: Data Augmentation Techniques for Tomato Dataset

| No. | Technique | Parameter Value |
|---|---|---|
| 1 | Horizontal Flip | $p = 1.0$ |
| 2 | Vertical Flip | $p = 1.0$ |
| 3 | Rotation | $\pm 20°$ |
| 4 | Brightness Contrast | $\pm 0.2$ |
| 5 | Hue Saturation | H=$\pm$10, S=$\pm$15, V=$\pm$10 |
| 6 | CLAHE | Clip Limit = 3.0 |
| 7 | Gaussian Blur | $(3, 3)$ |

TABLE II: Statistics of the Tomato Dataset

| Dataset Category | Class | Original | Augmented |
|---|---|---|---|
| Tomato Dataset | Stage- 1 | 20 | 160 |
| | Stage- 2 | 20 | 160 |
| | Stage- 3 | 20 | 160 |
| | Stage- 4 | 20 | 160 |
| | Stage- 5 | 20 | 160 |

*2) Dragon Fruit Dataset:* Dragon fruit, a nutrient-rich tropical fruit. In this study, a publicly available dataset titled *Dragon Fruit Maturity Detection and Quality Grading Dataset*[19] was utilized. The dataset was collected from three locations in Bangladesh (Gazipur, Jhenaidah, and Munshiganj) during May 2023 to August 2023.

The dataset focuses on two primary tasks: maturity detection and quality grading. It consists of four classes: immature, mature, fresh, and defect dragon fruits.However,in this study, only the maturity detection subset (immature and mature classes) has been considered.

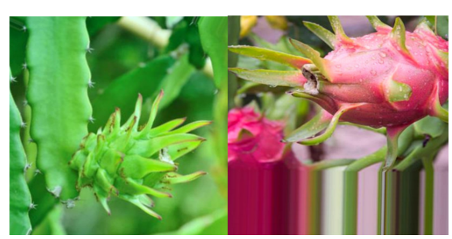

Fig. 3: Sample images from the dragon fruit dataset.

A total of 3779 original images were captured using high-resolution smartphone cameras (Redmi Note 11 Pro Plus and

Samsung S22). All images are stored in JPG format with a resolution of 800 $\times$ 800 pixels. Various data augmentation techniques were applied, including rotation, shifting, shearing, zooming, flipping and brightness adjustment. This resulted in a total of 10,010 augmented images.

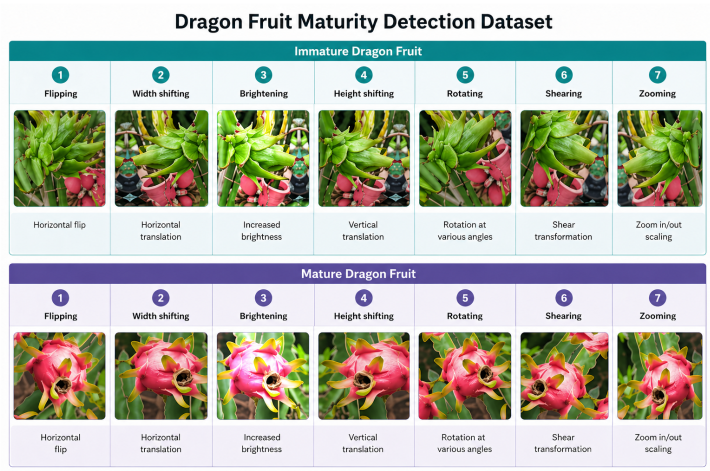


Fig. 4: Augumentation of images.

TABLE III: Data Augmentation Techniques for Dragon Fruit Dataset

| No. | Technique | Parameter Value |
|---|---|---|
| 1 | Horizontal Flip | Enabled |
| 2 | Rotation | $0°$ to $45°$ |
| 3 | Width Shift | 0.2 |
| 4 | Height Shift | 0.2 |
| 5 | Shearing | 0.2 |
| 6 | Zooming | 0.8 – 1.2 |
| 7 | Brightness Adjustment | 0.5 – 1.5 |
| 8 | Fill Mode | Reflect |

TABLE IV: Statistics of the Dragon Fruit Dataset

| Dataset Category | Class | Original | Augmented |
|---|---|---|---|
| Dragon Fruit Dataset | Immature | 1241 | 3000 |
| | Mature | 887 | 2010 |
| | Fresh | 898 | 2000 |
| | Defect | 754 | 3000 |
| Total | | 3779 | 10010 |

| **Hyperparameter** | **Value** |
|---|---|
| Image Size (IMG_SIZE) | 224 |
| Batch Size (BATCH_SIZE) | 32 |
| Initial Training Epochs (INITIAL_EPOCHS) | 10 |
| Fine-tuning Epochs (FINE_TUNE_EPOCHS) | 20 |
| Learning Rate (LEARNING_RATE) | $1 \times 10^{-4}$ |

### *B. Model Training*

The training process for models was performed in two stages to use transfer learning effectively.

**1) Base Training Stage:** Freezing of the model layers' base ensured the preservation of the pre-trained ImageNet weights. The training was limited to the classification layers, and the learning rate was set to $1 \times 10^{-4}$. This step allowed for the learning of new features specific to the task at hand without altering the pre-trained weights. The batch size was 32, and images were rescaled to have dimensions of 224$\times$224 pixels and normalized in the [0,1] scale. The model was trained for ten epochs, monitoring the validation loss for early stopping to avoid overfitting.

**2) Fine-Tuning Stage:** After the initial convergence, select convolutional blocks of the base network were unfrozen in order to tune the learned weights based on the target dataset. In the process, the training was continued for another 20 epochs while maintaining a low learning rate to guarantee stable weight updates. Finally, dropout layers were added in the fully connected layers to mitigate overfitting, while data augmentation was employed through rotation, flipping, and zooming operations.

For the optimization, the Adam optimizer was selected, along with the binary cross-entropy loss function, which typically performs well in binary classification problems. The validation parameters such as accuracy, precision, and recall were measured to monitor the effectiveness of the model training process. Also, the training and validation loss curves were tracked to ensure proper convergence.

### *C. Classification Models*

In this Study, we used four different pre-trained convoltion neural network models to decide maturity stages of tomato :EfficientNetV2B0 [15], MobileNetV2 [16], ConvNeXt-Tiny [17] and MicroViT-S1 [18].

**EfficientNetV2B0:** EfficientNetV2-B0 [15] is a convolutional neural network. It is built using three main types of blocks: an initial Conv 3×3 layer for basic feature extraction, followed by Fused-MBConv blocks in the early stages for fast and efficient computation, and MBConv blocks with Squeeze-and-Excitation (SE) in deeper layers for enhanced feature learning. The network concludes with a Convolution 1×1, global average pooling, and a fully connected output layer. EfficientNetV2-B0, pretrained on ImageNet, is utilized as an efficient backbone for crop maturity detection due to its fast fine-tuning and low computational cost. The lightweight model supports accurate and low-latency inference for real-time edge-based agricultural systems.

**MobileNetV2:** MobileNetV2 [16] is designed for efficient mobile/edge deployment, using inverted residual blocks with linear bottlenecks. Its structure can be divided into five main types of blocks/layers:Input Layer, Initial Convolution, Bottleneck Blocks (Inverted Residual + Linear Bottleneck), Final Convolution, Global Average Pooling + Fully Connected Layer. In MobileNetV2, a bottleneck block has 3 main layers:1×1 Pointwise Convolution, 3×3 Depthwise Convolution and 1×1 Pointwise Convolution. MobileNetV2 is used for crop maturity detection due to its lightweight architecture, efficient feature extraction capability, and feasibility for real-time deployment on agricultural edge devices.

**ConvNeXt-Tiny:**ConvNeXt-Tiny [17] is used as the backbone model for crop maturity classification. It is an improved CNN model based on the ResNet architecture and inspired by ideas from Vision Transformers. ConvNeXt-Tiny follows a 4-stage hierarchical structure with 18 ConvNeXt blocks. Its early layers capture low-level features such as edges and colors. and deeper layers capture high-level semantic features.

The model uses larger 7×7 depthwise convolutions, along with inverted bottleneck blocks, Layer Normalization, and GELU activation, to learn meaningful features more effectively and train more stably.

**MicroViT-S1:** MicroViT-S1 [18] is a lightweight Vision Transformer designed for efficient image classification. It adopts a multi-stage MetaFormer architecture constructed by stacking multiple MicroViT encoder blocks. The model consists of an initial patch embedding layer followed by 10 lightweight transformer encoder blocks arranged in four stages (2–2–4–2), and a final classification head. The overall architectural pipeline of MicroViT-S1 includes the following stages: input image preprocessing, patch embedding with positional encoding, lightweight token generation, transformer encoder stack, feature aggregation, and classification.

**Algorithm 1** Generalized Training Pipeline for Multi-Model Crop Maturity Detection

**Require:** Tomato dataset $D_T$, Dragon Fruit dataset $D_D$
**Require:** Model set $\mathcal{M}$ = {EfficientNetV2B0, MobileNetV2, ConvNeXt-Tiny, MicroViT-S1}
**Ensure:** Best trained models $\mathcal{M}^*$ for both crops
**function** TRAINANDEVALUATE($D$, model_type)
  Split dataset $D \to \{D_{train}, D_{val}, D_{test}\}$
  Assign labels based on dataset classes
  Initialize model $M \leftarrow$ model_type
  Replace final classification layer based on number of classes
  optimizer ← Adam($lr = 1 \times 10^{-4}$)
  loss ← CrossEntropyLoss()
  **/* Base Training */**
  Freeze backbone layers of $M$
  **for** epoch = 1 $\to E_1$ **do**
    Train($M, D_{train}$)
    Validate($M, D_{val}$)
    Save best model (highest validation accuracy)
  **end for**
  **/* Fine-Tuning */**
  Unfreeze selected layers of $M$
  **for** epoch = 1 $\to E_2$ **do**
    Train($M, D_{train}$)
    Validate($M, D_{val}$)
    Save best model
  **end for**
  Load best saved model
  Evaluate($M, D_{test}$) → {Accuracy, Precision, Recall, F1-score}
  **return** $M$
**end function**

**for** each dataset $D \in \{D_T, D_D\}$ **do**
  **for** each model $m \in \mathcal{M}$ **do**
    $M^*_{D,m} \leftarrow$ TRAINANDEVALUATE($D, m$)
  **end for**
**end for**
**return** $\mathcal{M}^*$

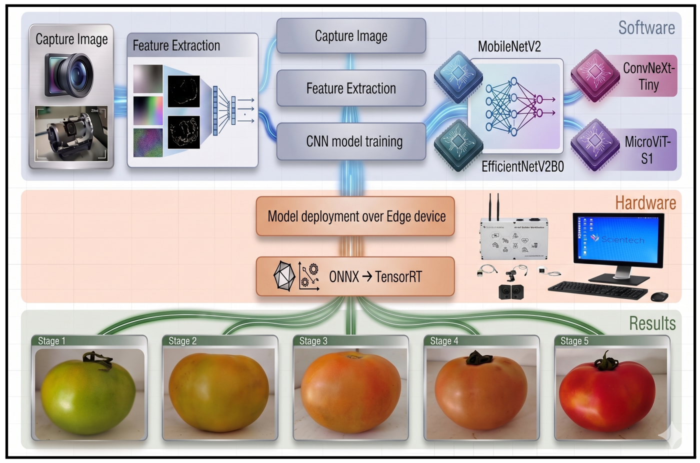


Fig. 5: Workflow

*D. Models Evaluation*

The models performance was assessed on previously unseen images taken directly from the Internet, ensuring evaluation on diverse and realistic data.


Fig. 6: Unseen Images of tomato and dragon fruit for performance evaluation

Fig. 6 shows the sample images of tomato and dragon fruit used for model training and evaluation. The images represent various maturity stages, including immature (green) and mature (red) fruits, [8]ensuring balanced learning across variations in color, texture, and shape. The figure also presents the predicted class labels, demonstrating ability of model's to accurately differentiate between mature and immature samples.

TABLE V: Evaluation Metrics

| Metric | Description | Formula |
|---|---|---|
| Accuracy | Overall correct predictions | $\frac{TP+TN}{TP+TN+FP+FN}$ |
| Precision | Correct positive predictions | $\frac{TP}{TP+FP}$ |
| Recall | Actual positives identified | $\frac{TP}{TP+FN}$ |
| F1-Score | Balance of precision and recall | $\frac{2\times P\times R}{P+R}$ |
| Confusion Matrix | Prediction results summary | $\begin{bmatrix} TP & FP \\ FN & TN \end{bmatrix}$ |

In order to assess the effectiveness of the models, various measures have been applied, which are detailed in Table V.

In table, $TP$ denotes true positives, $TN$ denotes true negative, $FP$ shows false positives , and $FN$ shows false negative.

### *E. Edge Device and Deployment Setup*

To examine the practical deployment capability of the proposed tomato maturity detection system, the selected trained models were deployed on the Scientech 6205AI Artificial Intelligence Workstation. This platform is intended for edge AI applications and provides a suitable environment for evaluating real-time deep learning inference under limited hardware resources.

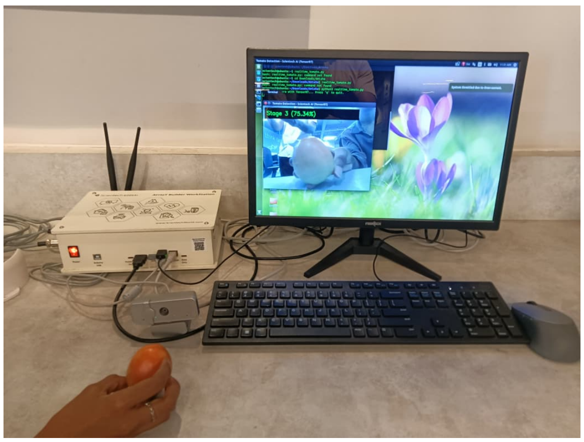

Fig. 7: Scientech 6205AI Artificial Intelligence Workstation

The Scientech 6205AI workstation is based on an embedded NVIDIA Jetson Nano architecture and is equipped with a quad-core ARM Cortex-A57 processor, a 128-core NVIDIA Maxwell GPU, and 4 GB LPDDR4 memory. Such a hardware configuration supports GPU-accelerated inference and is appropriate for lightweight and optimized computer vision models deployed at the edge.

The software environment used in this work included Ubuntu 18.04, JetPack R32.7.6, Python 3.6.9, and TensorRT 8.2.1. The deployment pipeline was developed using PyTorch, ONNX, TensorRT, and OpenCV. After model training, the selected models were exported and converted into optimized TensorRT inference engines to improve execution efficiency on the Scientech 6205AI platform.

The edge device was employed to evaluate deployment-oriented performance metrics such as throughput, latency, and GPU compute time. Since embedded systems provide limited memory and computational capability compared with conventional high-performance computing platforms, model optimization was necessary to achieve efficient real-time inference. Therefore, the Scientech 6205AI workstation served as an effective platform for validating the proposed AI-based agricultural application in an edge deployment setting.

## III. Results

The selected models were evaluated for tomato and dragon fruit maturity classification using accuracy, precision, recall, and F1-score. In addition, the tomato models were deployed on the Scientech 6205AI platform to examine their real-time inference behavior through throughput, mean latency, and GPU compute time.

### *A. Tomato Maturity Classification Results*

The tomato classification results are presented in Table V. Among the evaluated models, ConvNeXt-Tiny delivered the strongest overall performance and achieved perfect scores across all four evaluation metrics. MicroViT-S1 also performed at a very high level and remained close to ConvNeXt-Tiny in all measures. Although EfficientNetV2-B0 and MobileNetV2 produced slightly lower values, both models still showed reliable classification capability. These observations indicate that ConvNeXt-Tiny and MicroViT-S1 were the most effective models for tomato maturity recognition.

TABLE VI: Model Evaluation Matrics for Tomato

| Model | Accuracy(%) | Precision(%) | Recall(%) | F1 Score(%) |
|---|---|---|---|---|
| EfficientNetV2B0 | 97.50 | 97.54 | 97.50 | 97.49 |
| MobileNetV2 | 96.67 | 96.88 | 96.67 | 96.62 |
| ConvNeXt-Tiny | 100 | 100 | 100 | 100 |
| MicroViT-S1 | 99.17 | 99.20 | 99.17 | 99.17 |

### *B. Dragon Fruit Maturity Classification Results*

The dragon fruit results are summarized in Table VI. For this dataset, EfficientNetV2-B0, ConvNeXt-Tiny, and MicroViT-S1 achieved perfect classification performance. MobileNetV2 also produced strong results, with only a small reduction compared with the other models. This shows that all four models were suitable for dragon fruit maturity classification, while three of them achieved fully correct prediction on the evaluated samples.

TABLE VII: Models Evaluation Metrics for Dragon Fruit

| Model | Accuracy(%) | Precision(%) | Recall(%) | F1 Score(%) |
|---|---|---|---|---|
| EfficientNetV2B0 | 100 | 100 | 100 | 100 |
| MobileNetV2 | 99.07 | 99.21 | 98.88 | 99.03 |
| ConvNeXt-Tiny | 100 | 100 | 100 | 100 |
| MicroViT-S1 | 100 | 100 | 100 | 100 |

### *C. Edge Deployment Performance for Tomato Models*

To study deployment efficiency, the optimized TensorRT engines of the tomato models were executed on the Scientech 6205AI workstation. The results listed in Table VII show that MicroViT-S1 achieved the best real-time inference performance, combining the highest throughput with the lowest latency among the tested models. MobileNetV2 and EfficientNetV2-B0 also provided comparatively efficient runtime behavior, whereas ConvNeXt-Tiny required more inference time despite its strong classification accuracy. This comparison shows that the most accurate model is not always the most efficient in deployment. The small gap between latency and GPU compute time also indicates that the deployment pipeline was effectively optimized for edge execution.

TABLE VIII: Edge Deployment Performance for Tomato Models on Scientech 6205AI

| Model | Throughput (FPS) | Mean Latency (ms) | GPU Compute Time (ms) |
|---|---|---|---|
| EfficientNetV2-B0 | 51.10 | 21.00 | 18.00 |
| MobileNetV2 | 52.80 | 18.90 | 18.50 |
| ConvNeXt-Tiny | 45.23 | 24.00 | 27.12 |
| MicroViT-S1 | 61.73 | 16.14 | 16.00 |

## IV. Conclusion

This study proposes a deep learning framework to distinguish the maturity levels of tomato and dragon fruit from their images. Various models have been experimented with to analyze their efficiency concerning accuracy and usability aspects.

As seen from the results obtained, the ConvNeXt-Tiny architecture shows the highest level of accuracy in predicting the maturity of tomatoes and dragon fruit. However, as far as real-time applications are concerned, MicroViT-S1 offers better speed and lower latency in terms of execution.

Hence, it all comes down to the requirements of particular applications to choose between ConvNeXt-Tiny and MicroViT-S1 for detecting the maturity of crops from their images. Consequently, this study proves the potential of deep learning in detecting the crop maturity in an automatic manner without damaging them.

The proposed system can be enhanced further by using large amounts of diverse training data and integrating it with real-time farming tools such as IoT devices or assistance technologies.

## V. Acknowledgments

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## References

[1] A. B. S. M. I. J. N. P. N. S. N. J. R. P. S. K. V. Y. A. K. G. A. K. J. Anjali, Ankita Jena, “State-of-the-art non-destructive approaches for maturity index determination in fruits and vegetables: principles, applications, and future directions,” Food Production, Processing and Nutrition, vol. 6, 2024.

[2] N. Begum and M. K. Hazarika, “Maturity detection of tomatoes using transfer learning,” Measurement: Food, vol. 7, p. 100038, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S277227592200017X

[3] S. R. Nagesh Appe, G. Arulselvi, and G. Balaji, “Tomato ripeness detection and classification using vgg based cnn models,” International Journal of Intelligent Systems and Applications in Engineering, vol. 11, p. 296–302, Feb. 2023.

[4] Y. Mu, T.-S. Chen, S. Ninomiya, and W. Guo, “Intact detection of highly occluded immature tomatoes on plants using deep learning techniques,” Sensors, vol. 20, no. 10, 2020.

[5] M. Waseem, M. M. Sajjad, L. H. Naqvi, Y. Majeed, T. U. Rehman, and T. Nadeem, “Deep learning model for precise and rapid prediction of tomato maturity based on image recognition,” Food Physics, vol. 2, p. 100060, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2950069925000143

[6] H. S. Mputu, A. Abdel-Mawgood, A. Shimada, and M. S. Sayed, “Tomato quality classification based on transfer learning feature extraction and machine learning algorithm classifiers,” IEEE Access, vol. 12, pp. 8283–8295, 2024.

[7] H.-T. Vo, K. C. Mui, N. N. Thien, and P. P. Tien, “Automating tomato ripeness classification and counting with yolov9,” International Journal of Advanced Computer Science and Applications, vol. 15, no. 4, 2024.

[8] project cangp, “Fruitdetectionv3 (3 class) dataset,” apr 2023.

[9] A. Khan, T. Hassan, M. Shafay, I. Fahmy, N. Werghi, S. Mudigansalage, and I. Hussain, “Tomato maturity recognition with convolutional transformers,” Scientific Reports, vol. 13, no. 1, December 2023. [Online]. Available: http://dx.doi.org/10.1038/s41598-023-50129-w

[10] R. Trigubenko and hfujihara, “Laborotomato: Instance segmentation dataset,” 2020. [Online]. Available: https://github.com/laboroai/LaboroTomato#readme

[11] M. Afonso, H. Fonteijn, F. S. Fiorentin, D. Lensink, M. Mooij, N. Faber, G. Polder, and R. Wehrens, “Tomato fruit detection and counting in greenhouses using deep learning,” Frontiers in Plant Science, vol. 11, p. 571299, 2020. [Online]. Available: https://doi.org/10.3389/fpls.2020.571299

[12] S. B. Nikam, P. S. Kumar, S. B. Lande, V. J. Nagalkar, and G. C. Wakchaure, “Predictive Classification Model for Quality Grading and Maturity Detection of Dragon Fruit Using Fused Deep CNN Feature and Ensemble Learning.”

[13] A. N, D. V. Shah, D. P, and R. Niharika, “Dragon Fruit Maturity Detection and Quality Grading with Scalable Deep Neural Networks,” in Proc. 2025 3rd Int. Conf. Intelligent Data Communication Technologies and Internet of Things (IDCIoT), 2025.

[14] T. Khatun, A. Razzak, M. S. Islam, and M. S. Uddin, “An extensive real-world in-field tomato image dataset involving maturity classification and recognition of fresh and defect tomatoes,” Department of Computer Science and Engineering, Daffodil International University, Dhaka, Bangladesh.

[15] M. Tan and Q. Le, “Efficientnetv2: Smaller models and faster training,” in Proceedings of the 38th International Conference on Machine Learning, ser. Proceedings of Machine Learning Research, M. Meila and T. Zhang, Eds., vol. 139. PMLR, 18–24 Jul 2021, pp. 10 096–10 106.

[16] M. Sandler, A. G. Howard, M. Zhu, A. Zhmoginov, and L. Chen, “Mobilenetv2: Inverted residuals and linear bottlenecks,” in 2018 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2018, Salt Lake City, UT, USA, June 18-22, 2018. Computer Vision Foundation / IEEE Computer Society, 2018, pp. 4510–4520.

[17] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, “Aconvnet for the 2020s,” in 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, pp. 11 966–11 976.

[18] N. Setyawan, C.-C. Sun, M.-H. Hsu, W.-K. Kuo, and J.-W. Hsieh, “Microvit: A vision transformer with low complexity self attention for edge device,” in 2025 IEEE International Symposium on Circuits and Systems (ISCAS), 2025, pp. 1–5.

[19] T. Khatun, M. A. S. Nirob, P. Bishshash, M. Akter, and M. S. Uddin,“A comprehensive dragon fruit image dataset for detecting the maturity and quality grading of dragon fruit,” Data in Brief, vol. 52, p. 109936, 2024..